Sec 10.4 Motion Through Space

You might recall that the derivative of a position function is velocity and that the derivative of a velocity function is the acceleration. Mathematical arguments aside, this is simply because the rate of change of position with time is velocity and the rate of change of velocity with time is acceleration. In this way we can say:

If the position of a moving object is given by a vector function:

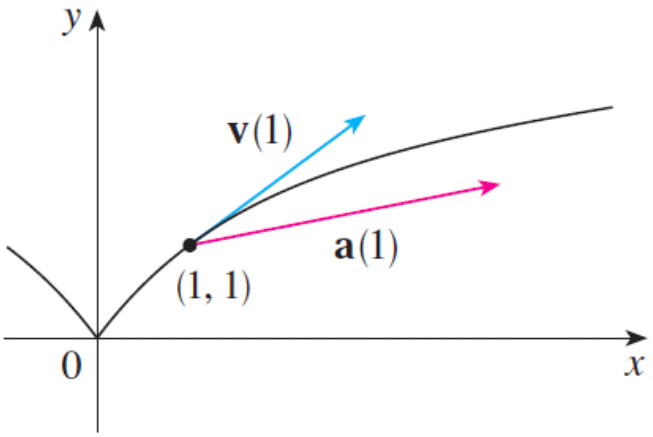
$\vec{r}(t) =< f(t), g(t), h(t) >$

Then the velocity of the particle is given by:

$\vec{v}(t) = \dfrac{d\vec{r}(t)}{dt} =< f'(t), g'(t), h'(t) >$ The Speed of the particle is therefore given by: $|\vec{v}(t)|$

Its acceleration is given by:

$\vec{a}(t) = \dfrac{d\vec{v}(t)}{dt} = \dfrac{d^2\vec{r}(t)}{dt} =< f''(t), g''(t), h''(t) >$



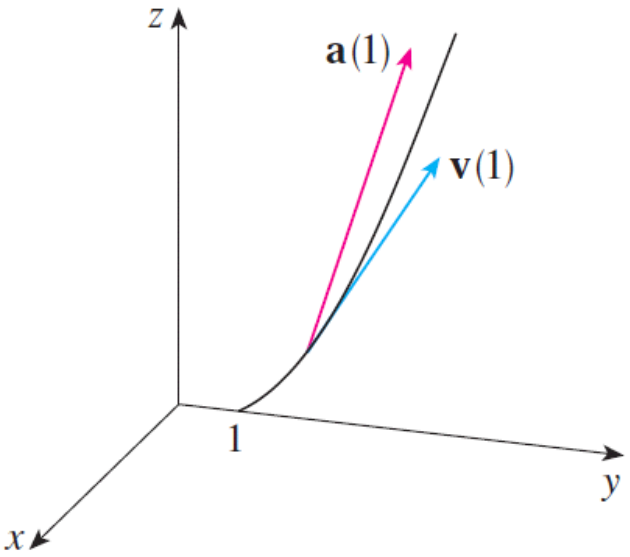


FIGURE 2

Sec 10.4 Motion Through Space

HW Example

3–8 Find the velocity, acceleration, and speed of a particle with the given position function. Sketch the path of the particle and draw the velocity and acceleration vectors for the specified value of t.

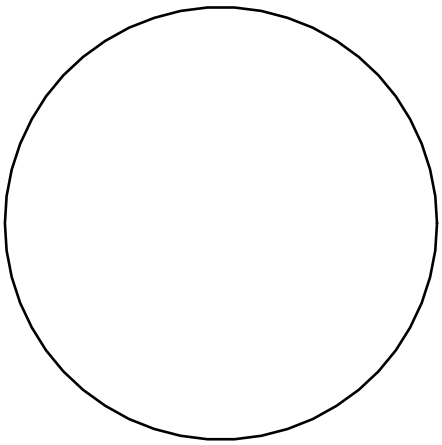
7. $\mathbf{r}(t) = t\,\mathbf{i} + t^2\mathbf{j} + 2\,\mathbf{k}, \quad t = 1$

Sec 10.4 Motion Through Space

Uniform Circular Motion

Suppose we have a particle moving in a circle with position vector given by $\vec{R}(t) < \mathrm{rcos}(\omega t)\,, \mathrm{rsin}(\omega \mathrm{t}) >$

What is the force acting on the particle?



Sec 10.4 Motion Through Space

Finding position given acceleration. Hw Example

15–16

(a) Find the position vector of a particle that has the given acceleration and the specified initial velocity and position.

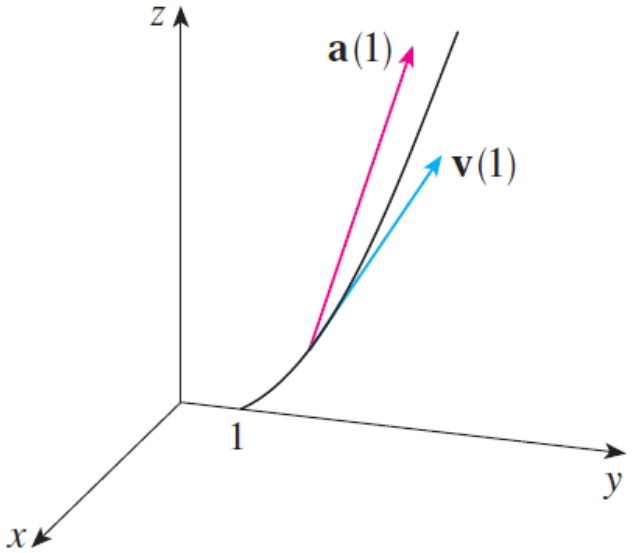


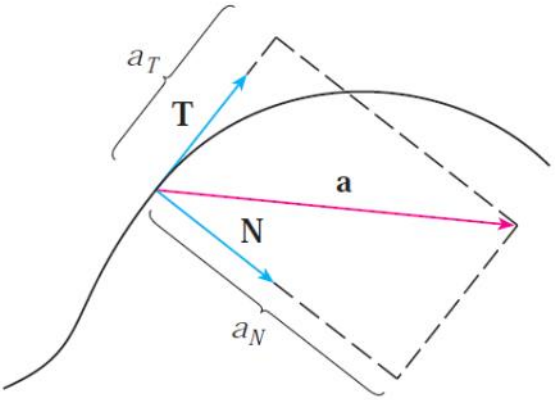
(b) Use a computer to graph the path of the particle.

15. $\mathbf{a}(t) = 2t\,\mathbf{i} + \sin t\,\mathbf{j} + \cos 2t\,\mathbf{k}, \quad \mathbf{v}(0) = \mathbf{i}, \quad \mathbf{r}(0) = \mathbf{j}$

Sec 10.4 Motion Through Space

Tangential and Normal components of acceleration.





$$\vec{a} = a_T\vec{T} + a_N\vec{N}$$

What are a_T and a_N ? How do we find them? We need to decompose the acceleration vector into a sum of two orghognal vectors.....

We can do this using our definitions of the Tangential and Normal vectors and Curvature.
And recalling this rule for scalar times vector function differentiation

3. $\dfrac{d}{dt}[\,f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} = \frac{\mathbf{v}(t)}{|\mathbf{v}(t)|} = \frac{\mathbf{v}}{v} \qquad \mathbf{N}(t) = \frac{\mathbf{T}'(t)}{|\mathbf{T}'(t)|} \qquad \kappa = \frac{|\mathbf{T}'|}{|\mathbf{r}'|} = \frac{|\mathbf{T}'|}{v} \quad \text{so} \quad |\mathbf{T}'| = \kappa v$$

$$\mathbf{v} = v\mathbf{T}$$

Sec 10.4 Motion Through Space

So if we decompose the acceleration vector into a sum of two orthogonal vectors we have:

$$\mathbf{a} = v'\mathbf{T} + \kappa v^2\mathbf{N}$$

Where $v = |\boldsymbol{v}(\boldsymbol{t})|$ is the speed, **T** the unit tangent vector, κ the curvature and **N** the unit normal vector.

We can also write this as:

$$\mathbf{a} = a_T\mathbf{T} + a_N\mathbf{N}$$

$$a_T = v' \qquad \text{and} \qquad a_N = \kappa v^2$$

One can also show that:

$$a_T = v' = \frac{\mathbf{v} \cdot \mathbf{a}}{v} = \frac{\mathbf{r}'(t) \cdot \mathbf{r}''(t)}{|\mathbf{r}'(t)|}$$

and

$$a_N = \kappa v^2 = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|^3}|\mathbf{r}'(t)|^2 = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|}$$

Sec 10.4 Motion Through Space

$$\mathbf{a} = a_T\mathbf{T} + a_N\mathbf{N}$$

HW Example

33–36 Find the tangential and normal components of the acceleration vector.

$$a_T = v' \qquad \text{and} \qquad a_N = \kappa v^2$$

35. $\mathbf{r}(t) = \cos t\,\mathbf{i} + \sin t\,\mathbf{j} + t\,\mathbf{k}$

Sec 10.4 Motion Through Space

Back to that circular motion…… We found that the acceleration on a particle traveling in a circle was given by: $\vec{a}(t) = r\omega^2 < \mathrm{rcos}(\omega t)\,, \mathrm{rsin}(\omega t) > = \mathrm{r}\omega^2\vec{\mathrm{R}}(\mathrm{t})$

Let's decompose this into the normal and tangential components… We Know:

$$\mathbf{a} = v'\mathbf{T} + \kappa v^2\mathbf{N}$$

Sec 10.4 Motion Through Space

Back to that roller coaster… Recall from the last lab we had a roller coaster moving down a spiral section of a track with position vector $\vec{r} = < 10\sin(\pi t), 10\cos(\pi t), 100 - 10t >$ Find The force on the coster car as it moves down the track.